

Results

77 out of 102 patients were treated with interstitial implants (no. of implants ranging 3-21, median of 6) and 25 by a mold technique. The 5-year control rates were 95%, 91% and 83% for local, regional and local-regional control respectively. Since the introduction of image guided brachytherapy in 2010, 5-year local control was 100% ($n = 42$). Tumor volume $\geq 2.3 \text{ cm}^3$ resulted in worse 3-year regional control (61%) compared to volume $< 2.3 \text{ cm}^3$ (96%; $p = .01$). Ultimate regional control after salvage treatment was 96%, with no significant difference between subgroups by tumor volume (92% for $\geq 2.3 \text{ cm}^3$ vs 96% for $< 2.3 \text{ cm}^3$; $p = .57$). Patient assessed cosmetic and functional satisfaction were both rated high (mean 3.7 and 4.0 out of 5 respectively).

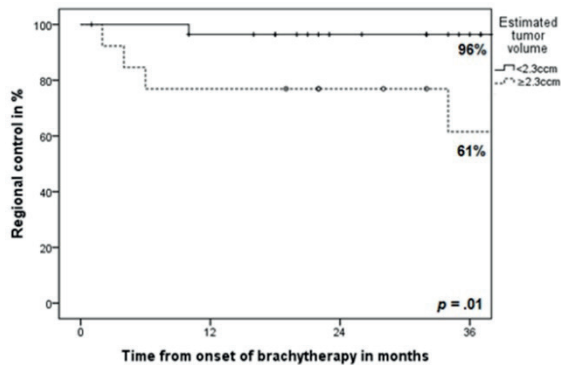


Figure 2: 3-year regional control by tumor volume (Log-rank test, $p = .01$).

Conclusion

We report the largest patient cohort treated with brachytherapy as the sole treatment for nasal vestibule carcinoma to date. Brachytherapy offers excellent local and regional control for Wang T1-T2 tumors with high patient satisfaction. Tumor volume is an adequate predictive factor for patients at risk of regional recurrence, but ultimate control rates after salvage treatment are high. Therefore, we do not recommend elective treatment of the neck. In future work, we want to explore sentinel node biopsy as diagnostic tool for high-risk patients with nasal vestibule carcinoma.

OC-0290 A predictive model for visus loss after uveal melanoma interventional radiotherapy (brachytherapy)

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Purpose or Objective

The aim of this study was to develop a predictive model on visus loss after ruthenium-106 plaque brachytherapy in uveal melanoma.

Material and Methods

Patients from institutional database with choroidal melanoma treated with ruthenium-106 plaque from December 2006 to November 2016 were selected. In each case the prescribed dose was 100 Gy at tumor apex. Factors analyzed were Age at diagnosis, Tumor thickness, Longitudinal tumor diameter, Transversal tumor diameter, Distance to optic disc, Gender, Quadrant, Localization, Morphology, Pigmentation, Dose fovea, Dose disco, Distance to the fovea, Diabetes, Hypertension. Univariate statistical tests (Pearson's Chi square and Two Sample t-test) and multivariate logistic regression were used to define the impact of baseline patient factors on the visus loss. A p-value ≤ 0.05 was considered significant. The model was chosen through stepwise selection on significative features at univariate analysis, and its performance was evaluated with internal cross-validation using Area Under the ROC Curve (AUC) and calibration with Hosmer-Lemeshow test.

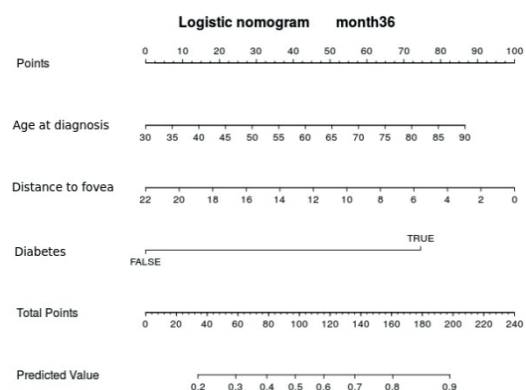
Results

187 patients with a median age of 65 years (range: 17-87) were considered for this analysis. The median follow-up was 72.25 months. Of 187 patients, 93% were alive. Visus loss was found in 98 patients (52.4%) at 3 year after the treatment. Distance to fovea was the main prognostic factor of the predictive model (odds ratio of 0.905 [0.845-0.965] $p = 1.5e-03$). Other prognostic factors were Age at diagnosis (odds ratio of 1.03 [1.01-1.05] $p = 0.025$) and Diabetes (odds ratio 5.10 [3.50-6.70] $p=0.039$) (table 1). Based on this analysis we developed a nomogram (fig.1), with an AUC of 0.73, useful in the clinical practice. The calibration show no statistical difference between actual and predicted 3-year visus loss ($p=0.28$).

Feature	Odds ratio	p-value
Age at diagnosis	1.03	0.025
Distance to fovea	0.905	1.5e-03
Diabetes	5.10	0.039

Conclusion

Our prognostication model could be a tool for predicting visus loss at 3 years after treatment. Indeed, this analysis revealed that distance to the fovea, age and diabetes can help to predict visus loss at 3 years after treatment: a predictive model is provided and reasonable performance (AUC) encourage further investigations along this direction.



OC-0291 3D image-guided treatment planning of Ru-106 brachytherapy for choroidal melanomas

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Purpose or Objective

The majority of current standard treatment procedures for Ruthenium-106 (Ru-106) brachytherapy for choroidal melanomas use only limited image guidance and no 3D treatment planning for plaque positioning or treatment time calculation. We evaluated the potential impact of introduction of 3D treatment planning in terms of tumour control probability (TCP) and normal tissue complication probability (NTCP).

Material and Methods

Retrospectively collected data from 90 consecutive patients with primary choroidal melanomas treated with Ru-106 plaque brachytherapy (from 2005-2008 at our institution) were used. All patients were originally treated to a prescribed dose of 100 Gy using an in-house developed spreadsheet based on depth dose data, tumour height measured from ultrasound B-scans, and activity of the plaque at the time of insertion. Data for all patients were recreated using the image-based 3D treatment planning software Plaque Simulator™ as well as pre- and posttreatment retinal images and treatment times obtained from patient records. Dose metrics for the tumour, the macula and the optic disc were extracted. The minimum tumour dose was related to tumour outcome while mean normal tissue doses were related to incidence of radiation induced maculopathy and optic neuropathy, using logistic regression to create TCP and NTCP models.

Treatment plans for a subset of 35 patients were subsequently re-optimised using Plaque Simulator™ and the original imaging. All optimised plans were approved by a consultant ophthalmologist. Optimisation of the plans was done in three steps; initially the treatment time was optimised such that the entire tumour received the prescribed dose, secondly the location of the plaque was changed to cover the tumour base as much as possible, and lastly the treatment time and plaque location were optimised concurrently. TCP and NTCP for the original and for the 3D optimised plans were compared using Wilcoxon signed rank test.

Results

The median tumour dose in the clinical plans was 73 Gy (IQR: 47-117) equivalent to TCP of 74 % (53-93), while median tumour dose in the fully optimised plans was 101 Gy (100-102) equivalent to TCP of 88 % (88-89).

The median increase in TCP was 14 % (IQR: -4-35, $p=0.0002$) with a small subset of patients having decreased TCP all due to original over dosage (tumour Dmin>100 Gy). The median increase in NTCP for maculopathy was 5 % (0-24, $p=0.0002$). The median increase in NTCP for optic neuropathy was 1 % (0-7, $p=0.01$).

Location of the plaque proved important to reduce total dose and thus spare normal tissues, while optimising treatment time was essential to achieve TCP.

Conclusion

3D planning allows for improved treatment delivery for Ru-106 brachytherapy of choroidal melanomas, resulting in a significant increase in expected tumour control. However, due to the close proximity of the macula and the optic disc, this may come at the cost of increased probability of normal tissue toxicity, even using 3D image guidance for treatment planning.

Proffered Papers: PH 5: MRI for treatment planning and delivery

OC-0292 Applicability of MR-only based radiation therapy treatment planning for intracranial target volumes

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Purpose or Objective

Radiation therapy treatment planning for intracranial target volumes is usually performed on fused MRI and CT data. While it is desirable to contour organs at risk (OAR) and target volumes (GTV) on MRI data for soft tissue contrast, the electron density information of the CT is required to determine the absorbed dose. With newly available software and MR sequences, as well as technical improvements like homogenous field gradients in currently available MR-scanners, it recently became feasible to use solely MRI for contouring and dose calculation. For this purpose the MR-data is converted into a synthetic-CT (sCT) with electron densities similar to a treatment-planning-CT (pCT). We compared sCT and pCT with respect to: (a) the dosimetric differences, (b) the 2D-positioning accuracy of image guided radiation therapy (IGRT) with digitally reconstructed radiographs (DRR), and (c) the 3D-IGRT positioning accuracy with a cone-beam-CT (CBCT).

Material and Methods

In this study, in parallel to the regular treatment planning routine of using a diagnostic MRI for OAR and GTV delineation and a pCT for dose calculation, we performed an additional MRI scan (sequences: T1 Dixon vibe, T2 Space, PETRA, time-of flight sequences) for 15 patients with intracranial lesions. This MR-scan was performed in treatment position. For this purpose, a special in-house developed, MR-compatible flat couch surface was used on which all relevant positioning devices for radiation therapy treatments, such as a soft mask and a head cushion, were attached. Patients were scanned on a 3 T MR scanner (Magnetom Skyra, Siemens) using a Body18 coil. The MR-sequences were converted into a sCT with the Syngo.via Frontier syntheticCT prototype (Siemens). The resulting sCT were imported into a treatment planning system (Monaco, Elekta), the pCT based VMAT delivery sequences that were used for patient treatment were transferred to sCT, and the dose was recalculated. The resulting sCT dose distributions were compared to the ones on the pCT. Furthermore, DRR at 0° and 90° gantry angle were compared. CBCT (XVI, Elekta), obtained before the first treatment fraction, were fused with the corresponding sCT and pCT to evaluate possible positioning offsets.

Results

Dosimetric differences between sCT and pCT based plans were small. Differences in $D_{95\%}(PTV)$ and $D_{5\%}(PTV)$ were <1%. The maximum differences in organs at risk doses were less than 3.5%. Global gamma-analysis (1%, 1mm) had pass-rates above 99 % with a threshold of 25% of the max. dose. Image quality of DRR was sufficient for 2D-patient positioning. The registration offset between pCT and sCT on the first treatment fraction CBCT was 0.1 ± 0.2 , 0.0 ± 0.5 , and -0.1 ± 0.7 millimeter for all three space-directions.